## Aim

To perform exploratory data analysis (EDA) on a student performance dataset using Pandas and Matplotlib.

# Dataset

Create a CSV file named student_data.csv with the following data:

| RollNo | Name | Gender | Attendance | InternalMarks | ExternalMarks |
|---|---|---|---|---|---|
| 101 | Ravi | M | 85 | 32 | 55 |
| 102 | Priya | F | 92 | 38 | 58 |
| 103 | Kiran | M | 78 | 28 | 45 |
| 104 | Anu | F | 95 | 40 | 60 |
| 105 | Sita | F | 88 | 35 | 54 |
| 106 | Raj | M | 70 | 25 | 40 |
| 107 | Divya | F | 90 | 37 | 57 |
| 108 | Arun | M | 82 | 30 | 48 |
| 109 | Keerthi | F | 96 | 39 | 59 |
| 110 | Naveen | M | 75 | 27 | 44 |

## Tasks

Task A: Dataset Loading and Inspection

- Import the necessary Python libraries.
- Load the dataset into a Pandas DataFrame.
- Display the first five records.
- Display the last five records.
- Find the total number of rows and columns.
- Display the names of all columns.
- Display the data type of each column.
- Generate dataset information using info().

Task B: Statistical Analysis

- Calculate the mean attendance of all students.
- Calculate the average Internal Marks.
- Calculate the average External Marks.
- Find the maximum Internal Marks.
- Find the minimum Internal Marks.
- Find the maximum External Marks.

- Find the minimum External Marks.
- Calculate the standard deviation of Internal Marks.
- Generate summary statistics using describe().

Task C: Data Manipulation

- Create a new column named TotalMarks.
- TotalMarks=InternalMarks+ExternalMarks
- Create a new column named Percentage.
    1. Percentage=TotalMarks/100×100
- Display the updated DataFrame.

Task D: Filtering and Searching

- Display the details of students whose attendance is greater than 90%.
- Display the details of students whose TotalMarks exceed 90.
- Display the details of female students.
- Display the details of male students.
- Display students whose ExternalMarks are less than 50.

Task E: Ranking and Comparison

- Identify the student with the highest TotalMarks.
- Identify the student with the lowest TotalMarks.
- Display the top three performers based on TotalMarks.
- Display the bottom three performers based on TotalMarks.
- Determine how many students scored above the class average.

Task F: Group-wise Analysis

- Calculate the average TotalMarks of male students.
- Calculate the average TotalMarks of female students.
- Compare the average attendance of male and female students.

Task G: Data Visualization

- Plot a bar chart showing TotalMarks of all students.
- Plot a histogram of Attendance.
- Plot a pie chart showing the distribution of male and female students.
- Plot a scatter plot between Attendance and TotalMarks.
- Interpret the relationship between Attendance and TotalMarks.

Task H: Observations

- Write any three observations based on the analysis performed.
- State whether higher attendance appears to be associated with better academic performance.